Understanding GIS: Assessment 1 Report

[Subject: GEOG71551]
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The purpose of this assessment was to develop an algorithm to find and identify the shortest international border in the world and plot a map of it which involved the use of Natural Earth dataset (`ne_10m_admin_0_countries.shp`), ultimately identifying Italy and Vatican as the Shortest International border of 0.43km. For this, a shared boundary between the 2 countries was important and excluding the point touches and countries sharing no border.
The workflow of the assessment consisted of many central elements of the spatial data analysis like geometry processing, spatial indexing, etc.

The global shapefile data is loaded in the GeoDataFrame and to do the calculation efficiently STRtree spatial index and Geod geodesic calculator was used so that the computational cost is reduced to a significant amount and only geometrically relevant neighbours be calculated whose bounding boxes overlap, rather that testing every possible country which made the algorithm quite scalable for global dataset inspite having the large number of polygons.

The intersection code (`geometry.intersection()`) identifies shared boundary and the borders are calculate don ellipsoidal surface as the Euclidean distance would understate the true border length, hence WGS84 was used as for global mapping it provides an accurate and consistent Earth model for measuring distance. This computes topological intersection of two polygon boundaries which returns:

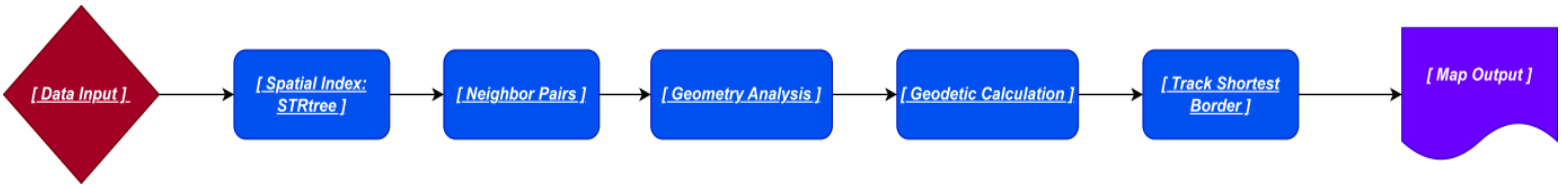
- **Empty** if borders don't touch
- **Point** if borders only touch at a corner
- **LineString/MultiLineString** if borders actually overlap (valid border)

In addition, we avoid the computation of countries A-B twice by using the condition `if idx_a >= idx_b: continue` so we don't get duplicate results and by using the `float('inf')` as intial minimum ensures any border that is real will be smaller and will be updated in shortest_length and best_pair. Borders can be stored as one continuous line (LineString) or in segments (MultiLineString). The cumulative segment distance represents the border length and by storing the shortest border found in (best_pair), the function repeatedly narrows down to Min. border length globally.

The final part of the assessment was to visualize the result in an effective map layout for which **Matplotlib** and **GeoPandas Was** used for layer entering**.** A crucial part was to choose the CRS for map production. Although CRS **EPSG:4326** Was used for distance calculations But reprojecting the data into a **Lambert conformal conic (LCC)** projection was required to visualize. LCC was chosen as it is ideal for regions between **20°N and 60°N latitude** where Italy and Vatican lie And also minimizes the angular distortion And visually balanced projection for small regions. Also **EPSG:32633** could be used here but it was not required as not much of numerical precision was required for the task appointed. Similarly, **Web Mercator** could be used but would have a severe distortion at these latitudes as it is not for cartographic accuracy.

Finally to visualize the key spatial elements Different color schemes were adopted For example the countries were displayed using soft but contrasting colors without creating a distraction and the border with a bold purple color in order to emphasize it as the focal element. To improve interpretability scale bar, legend and north arrow were used. Clarity and meaning of the map was the priority over the decoration.

Despite the interpretability several limitations were also there like the Natural Earth data set is quite generalized and deviates from total accuracy. Also the borders that are represented as simple polygons do not fully account for topography in the real world As some borders are politically contested (like Kashmir) but the data only recognizes the international boundaries. The final map successfully communicates the shortest international border . The solution successfully identified the border by combining spatial indexing, geodetic distance calculation and topological geometry to produce a result that was accurate and easy to interpret due to the visual hierarchy, proper projection and cartographic balance. The final Result is represented in **Fig. 2**.



[Fig. 1 represents the workflow followed throughout the assessment]

[Fig 2. Final Result]

